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Device for generation and distribution of electrical power

Abstract

A device for generation and distribution of electrical power has an electric motor which provides kinetic power to an electric generator. The electric motor is powered by a plurality of batteries which are operably connected to it. The electric generator provides power to a series of charge controllers and inverters which can provide power either to recharge the plurality of batteries and/or to the load source. A first circuit has the plurality of batteries powering the electric motor to provide kinetic power to the electric generator. The second circuit can be used to either recharge the plurality of batteries or provide power to the electric motor to provide kinetic power to the electric generator.

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Background/Summary

(1) This invention relates to a device for generation and distribution of electrical power, and more particularly, to a device for generation and distribution of electrical power which generates electrical power and then, distributes it for use for household/building purposes or for distribution to the electrical power grid.

BACKGROUND OF THE INVENTION

(2) Houses or other buildings (hereinafter “houses”) require electrical power for functioning of many important components or appliances. Generation of electrical power is important for houses which are off the electrical power grid. These houses still need electrical power, but may not have access to it due to remoteness of a location or due to an emergency situation such as a hurricane, earthquake, tornado, flooding, or other shut down of the electrical power grid. Providing access to power to houses in these locations or these situations through an apparatus that generates and distributes electrical power will be a useful invention.

(3) Electrical power demand is increasing. There is increased demand due to the charging of electric vehicles and a technology-based society which requires electrical power to operate computers and other electronic devices. An apparatus that generates and distributes electrical power will be a useful invention.

SUMMARY OF THE INVENTION

(4) An objective of the present invention is the provision of a device for generation and distribution of electrical power which can generate and distribute electrical power independently of the electrical power grid.

(5) Another objective of the present invention is the provision of a device for generation and distribution of electrical power which can generate and distribute electrical power to a house that is not connected to the electrical power grid.

(6) A still further objective of the present invention is the provision of a device for generation of electrical power that can generate and distribute electrical power to houses in an emergency situation where power from the electrical power grid is not available.

(7) Other objectives and advantages of the invention will become apparent from the description of the preferred embodiments herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 depicts a schematic diagram of the device for generation and distribution of electrical power **100** of this invention.
- (2) FIG. 2 depicts a schematic diagram of the device for generation and distribution of electrical power **100** of this invention.
- (3) Throughout the figures of the drawings, where the same part appears in more than one figure of the drawings, the same number is applied thereto.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (4) Reference will now be made in detail to several embodiments of the invention that are illustrated in accompanying drawings. Whenever possible, the same or similar reference numerals are used in the drawings and the description to refer to the same or like parts or steps. The drawings are in simplified form and are not to precise scale. For purposes of convenience and clarity only, directional terms such as top, bottom, left, right, up, down, over, above, below, beneath, rear, and front, can be used with respect to the drawings. These and similar directional terms are not to be construed to limit the scope of the invention in any manner. The words attach, connect, couple, and similar terms with their inflectional morphemes do not necessarily denote direct or intermediate connections, but can also include connections through mediate elements or devices.
- (5) All measures of voltage and current in the following description of the preferred embodiment are meant to be exemplary and not limit the disclosure in any manner.
- (6) Throughout the drawings, the term 4 Con. wire is used. This is an abbreviation for 4 control wires and while only one wire is shown in the illustration, there are in actuality four different wires connected between the same two points as the depicted wire.
- (7) Now adding FIG. 1 and FIG. 2 to the consideration, the structure and function of device **100** can be clearly seen. Device **100** has first circuit **102** and second circuit **104**. Device **100** also has a control panel **106**. Control panel **106** is connected to outlet **178** (which is preferably, but not required, a 120 volt, alternating current outlet) which is connected to breaker panel **172** (sometimes referred to as a “first breaker panel”) and breaker panel **172** provides electrical power (hereinafter “power”) to control panel **106**. Control panel **106** monitors and controls front inverter **160**, inverters and charge controllers **162**, **164**, **166**, and **168**, batteries **142**, electric generator **120**, and motor controller **130**. Control panel **106** monitors the function of device **100** and has a cat 5 data connection and Wi-Fi connection (or any other suitable internet connection) to allow internet access by an electronic device (phone, computer, tablet, or any other suitable electronic device) for the monitoring and controlling of device **100**.

First Circuit **102**

- (8) Device **100** is started by powering on/off switch connected to safety breaker **108** (not shown) and batteries **142** of battery bank **140** have an internal switch to activate them. Batteries **142** of battery bank **140** are connected in parallel and provide a constant flow of energy to start the device **100** and also store energy for future use. Battery bank **140** has battery management system **150**. Battery management system **150** protects the batteries **142** from over-charging. Battery management system **150** allows a charge at 48 volts direct current (hereinafter “VDC”), with up to 200 amperes (hereinafter “amps”) and a discharge at 48 VDC with up to 200 amps drawing or discharging at a time per battery **142** while receiving up to a charge of 48 VDC and 200 amps per battery **142** to charge battery bank **140** and to operate device **100**.
- (9) Battery bank **140** is preferably a graphene super conductor battery bank that consists of four batteries **142** at 48 VDC, 100 amps minimum with 200 amps maximum. However, any type of batteries **142** can be used and up to nine batteries can be used.
- (10) Once running, device **100** is powered by charge controllers and inverters **162**, **164**, **166**, and **168** (of the second circuit **104**). Once charge controllers and inverters **162**, **164**, **166**, and **168** take over powering device **100**, batteries **142** of battery bank **140** standby as a back-up power source for

device **100**.

(11) Power from batteries **142** of battery bank **140** flows through positively charged wires through safety breaker **108** at 48 VDC, 100 amps minimum and up to 200 amps maximum to a positively charged battery bus bar **144**. Power from batteries **142** also flows through negatively charged wires (four control wires are depicted in FIG. 1 as denoted as 4 Con. wire) at 48 VDC, 100 amps minimum and up to 200 amps maximum to relay control **190**. From relay control **190** power flows at 48 VDC, 100 amps minimum and up to 200 amps maximum to a negatively charged battery bus bar **144** (positively and negatively charged bus bars **144** are sometimes referred to as “first set of bus bars”). Battery bank **140** has a two battery **142** minimum, but can have up to nine batteries **142**. (12) Relay control **190** monitors battery bank **140**. Relay control **190** can turn on and off the charging of batteries **142** based on current power storage. Safety breaker **108** can cut off power from the batteries **142** to the remainder of device **100** in case of an emergency, natural disaster, or failure of device **100**.

(13) From positively and negatively charged battery bus bars **144**, power flows to front inverter **160** at 48 VDC, 100 amps minimum and up to 200 amps maximum. Front inverter **160** inverts power from 48 VDC to 240 volts alternating current (hereinafter “VAC”) (L1 and L2 exiting from front inverter **160** are connected in parallel so the combination of each at 120 VAC creates 240 VAC). Power flows to a breaker panel **172** (which is preferably a 60 amp breaker panel) at 240 VAC and 100 amps minimum and up to 200 amps maximum. From breaker panel **172**, power can flow to a transfer switch **101** (sometimes referred to as “first transfer switch”) at 240 VAC, 100 amps minimum and up to 200 amps maximum. Transfer switch **101** disconnects power from the power grid to the load source **210** and allows electric generator **120** to operably connect to and power the load source **210** which can be the home or other building, appliance, or other power receiver (hereinafter “load source”). The P1, P2, and P3 lines connect to the breaker panel or sub-panel for the load source **210** and power transfers to the breaker panel or sub-panel for the load source **210** at 240 VAC, 100 amps minimum and up to 200 amps maximum.

(14) Transfer switch **101** can also transfer power not used by the load source **210** to the power grid at 240 VAC, 100 amps minimum and 200 amps maximum.

(15) Power from breaker panel **172** can also be transferred to transfer switch **200** (sometimes referred to as “second transfer switch”) at 240 VAC, 100 amps maximum. Transfer switch **200** can transfer power to motor controller **130** at 240 VAC with a maximum of 100 amps. Motor controller **130** converts the 240 VAC to triple phase. From motor controller **130**, power can be transferred in triple phase at 360 VAC to electric motor **110**. Electric motor **110** rotates and converts electrical power into kinetic energy which operates electric generator **120** which converts kinetic energy back into electrical power. Power flows from electric generator **120** at up to a maximum of 450 VDC and is transferred through positively charged wire to positive bus bar **146** and through negatively charged wire to negative bus bar **146** (sometimes referred to as the “second set of bus bars”).

(16) It is preferred, but not required, that electric motor **110** is a three phase electric motor. However, any suitable electric motor **110** can be used and is encompassed by this disclosure.

(17) Electric motor **110** which is preferably a 15 horsepower electric motor, but can range between 5 horsepower and 30 horsepower, three phase with an operational 3000 revolutions per minute (RPM). The RPM of electric motor **110** can range between 1000 RPM and 3000 RPM.

(18) Electric generator **120** is preferably a 30 kilowatt (KW) permanent magnet generator, but can be up to a 50 KW permanent magnet generator. Any suitable generator can be used and all suitable generators are encompassed by this disclosure.

Second Circuit **104**

(19) Power from positive and negative bus bars **146** flows to charge controller and inverters **162**, **164**, **166**, and **168** at up to 450 VDC (as denoted by the letter A in FIG. 1 and FIG. 2). Four positively charged control wires exit from the positively charged bus bar **146** and four negatively charged control wires exit from the negatively charged bus bar **146** (as depicted by 4 Con. wire in

FIG. 2). Charge controller and inverters **162**, **164**, **166**, and **168** are connected in parallel and receive up to 450 VDC, and a maximum of 25 amps per charge controller and inverter so 100 amps for the four charge controllers and inverters **162**, **164**, **166**, and **168** in the depicted embodiment. Charge controller and inverters **162**, **164**, **166**, and **168** convert power to 48 VDC, 100 amp per charge controller and inverter. In the depicted embodiment four charge controllers **162**, **164**, **166**, and **168** are shown, but up to nine charge controllers and inverters can be utilized and are encompassed by this disclosure.

(20) Charge controllers and inverters **162**, **164**, **166**, and **168** have four wires exiting from each of them. Power flows from each charge controller and inverter **162**, **164**, **166**, and **168** at 240 VAC, 25 amp per **162**, **164**, **166**, and **168** so 100 amp output in combination to neutral (N) bus bar **192**, to bus bar **194** which is the ground (G), and to charger controller and inverter breaker panels **180**, **182**, **184**, and **186** through line 1 and line 2 (and if the maximum number of nine charge controllers and inverters are used they, in combination, put out 240 VAC, and 225 amps). Breaker panels **180**, **182**, **184**, and **186** can be individual breaker panels as shown or can be four independent sub-panels contained within one breaker box.

(21) From breaker panels **180**, **182**, **184**, and **186**, power flows from line 1 to line 1 bus bar **152** at 120 VAC, and 25 amps per charge controller and inverter **162**, **164**, **166**, and **168** so 100 amps. From breaker panels **180**, **182**, **184**, and **186** power flows from line 2 at 120 VAC and 25 amps per charge controller and inverter **162**, **164**, **166**, and **168** so 100 amps to line 2 bus bar **152** (the line 1 and line 2 bus bars **152** are sometimes referred to as the “third set of bus bars There is a line 1 and line 2 coming from each breaker panel **180**, **182**, **184**, and **186** (as indicated by 4 Con. wire in FIG. 2).

(22) From the L1 and L2 legs of bus bars **152**, power is transferred at 240 VAC, 100 to 225 amps to breaker panel **202** (sometimes referred to as a “second breaker panel”). Power from breaker panel **202** flows at 240 VAC, 100 to 225 amps to transfer switch **204** (sometimes referred to as “third transfer switch”).

(23) Transfer switch **204** disconnects power from the power grid to the load source **210** and allows electric generator **120** to connect to and power the load source **210**. The P1, P2, and P3 lines connect to the breaker panel or sub-panel for the load source **210** and power transfers to the breaker panel or sub-panel for the load source at 240 VAC, 100 amps minimum and up to 200 amps maximum.

(24) Transfer switch **204** (sometimes referred to as “third transfer switch”) can also transfer power not used by the load source **210** to the power grid at 240 VAC, 100 amps minimum and 200 amps maximum.

(25) From each charge controller and inverter **162**, **164**, **166**, and **168** a second positive and negative wire exit and connect to the positive and negative bus bars **144** (as denoted by B in FIG. 1 and FIG. 2). Power is transferred at 48 VDC, 100 amps minimum to 200 amps maximum to the positive and negative bus bars **144**. Each charge controller and inverter **162**, **164**, **166**, and **168** has a positive and negative wire exiting from it, four control wires enter each the positive bus bar **144** and the negative bus bar **144** (as depicted as 4 Con. wire in FIG. 1). From the positive and negative bus bars **144**, power can be transferred at 48 VDC, with a minimum of 100 amps and up to 200 amps maximum to front inverter **160** which then travels through the rest of first circuit **102**.

(26) Or, power can flow from the positive bus bar **144** to the safety breaker **108** at 48 VDC, with a minimum of 100 amps and up to 200 amps maximum. Then power flows from the safety breaker **108** to the batteries **142** at 48 VDC, with a minimum of 100 amps and up to 200 amps maximum.

(27) Or, power can flow from the negative bus bar **144** to the relay control **190** at 48 VDC, with a minimum of 100 amps and up to 200 amps maximum. Then power flows from the relay control **190** to the batteries **142** at 48 VDC, with a minimum of 100 amps and up to 200 amps maximum. This allows the charge controllers and inverters **162**, **164**, **166**, and **168** the ability to charge batteries **142**. Relay control **190** monitors the batteries **142** and can divert the transfer of power

from the positive and negative bus bars **144** to the front inverter **160** when the batteries **142** are fully charged.

(28) Transfer switch **200** is connected to both the motor controller **130** and breaker panel **202** (See C on FIG. **1** and FIG. **2**). Transfer switch **200** allows one circuit **102** or **104** of system **100** to take over when manually flipped so that one circuit **102** or **104** of system **100** can provide power if the other side is disabled. In an emergency, transfer switch **200** is manually flipped. This allows a user to shut off either the first circuit **102** or the second circuit **104** if there is a problem in system **100**.

(29) Transfer switch **200** can be switched so that electric motor **110** is powered by the flow of power from front inverter **160** running off batteries **142** or transfer switch **200** can be switched so that electric motor **110** is powered by the charge controllers and inverters **162**, **164**, **166**, and **168**.

(30) While various embodiments and aspects of the present invention have been described above, it should be understood that they have been presented by way of example only, and not limitation. Thus, the breadth and scope of the present invention should not be limited by any of the above exemplary embodiments.

(31) This application—taken as a whole with the abstract, specification, claims, and drawings being combined—provides sufficient information for a person having ordinary skill in the art to practice the invention as disclosed herein. Any measures necessary to practice this invention are well within the skill of a person having ordinary skill in this art after that person has made a careful study of this disclosure.

(32) Because of this disclosure and solely because of this disclosure, modification of this device can become clear to a person having ordinary skill in this particular art. Such modifications are clearly covered by this disclosure.

Claims

1. A device for generation and distribution of electrical power comprising: a) the device having an electric motor that powers an electric generator, wherein the electric motor is controlled by a motor controller; b) a plurality of batteries being operatively connected to a front inverter, wherein the front inverter is operatively connected to the electric motor; c) a plurality of charge controllers and inverters being operatively connected to both the plurality of batteries and the electric motor, wherein the plurality of batteries can be charged by the plurality of charge controllers and inverters, further wherein the electric motor can be powered by the plurality of charge controllers and inverters; and d) the electric generator being operatively connected to the load source, wherein the electric generator is configured to supply power to the load source.

2. The device of claim 1 further comprising: a) the device having a first circuit and a second circuit; b) the first circuit comprising: i) the plurality of batteries operatively connected to the front inverter; ii) the front inverter operatively connected to a second transfer switch; iii) the second transfer switch electrically connected to the motor controller; iv) the motor controller electrically connected to the electric motor; v) the electric motor kinetically connected to the electric generator; c) the second circuit comprising: i) at least one positively charged and at least one negatively charged bus bar of a second set of bus bars connected to both the generator and the plurality of charge controllers and inverters; ii) each of the plurality of charge controllers and inverters operatively connected to a third set of bus bars; iii) each of the bus bars of the third set of bus bars electrically connected to a second breaker panel; iii) the second breaker panel electrically connected to both the second transfer switch and a third transfer switch; and d) the second transfer switch configured to be switched to send power to the first circuit to allow the electric motor to be powered by the plurality of batteries, wherein the second transfer switch can also be switched to the second circuit to allow the electric motor to be powered by the plurality of charge controllers and inverters.

3. The device of claim 2 further comprising: a) the first circuit having a safety breaker electrically

connected to the plurality of batteries and a positively charged bus bar of a first set of bus bars, wherein the safety breaker can cut power to or from the plurality of batteries; b) the first circuit having a relay control electrically connected to the plurality of batteries and a negatively charged bus bar of the first set of bus bars, wherein the relay control is configured to monitor a level of charge of the plurality of batteries; and c) the positively charged and negatively charged bus bars of the first set of bus bars being electrically connected to the front inverter.

4. The device of claim 3 further comprising: a) the front inverter of the first circuit being electrically connected to a first breaker panel; b) the first breaker panel being electrically connected to the second transfer switch, wherein if the second transfer switch is switched to send power from the plurality of batteries to the electric motor, the first breaker panel transfers power to the second transfer switch; and c) the first breaker panel being electrically connected to the first transfer switch, wherein the first breaker panel is configured to send electrical power to a load source.

5. The device of claim 3 further comprising: a) each of the plurality of charge controllers and inverters being electrically connected to the first set of bus bars.

6. The device of claim 4 further comprising: a) each of the plurality of charge controllers and inverters being electrically connected to the first set of bus bars.

7. The device of claim 3 further comprising: a) each of the plurality of charge controllers and inverters being connected to each other in parallel; b) each of the plurality of charge controllers and inverters being electrically connected to one of a plurality of charge controller and inverter breaker panels, wherein there is one charge controller and inverter breaker panel for each charge controller and inverter; and c) each charge controller and inverter breaker panel being electrically connected to the third set of bus bars.

8. The device of claim 4 further comprising: a) each of the plurality of charge controllers and inverters being connected to each other in parallel; b) each of the plurality of charge controllers and inverters being electrically connected to one of a plurality of charge controller and inverter breaker panels, wherein there is one charge controller and inverter breaker panel for each charge controller and inverter; and c) each charge controller and inverter breaker panel being electrically connected to the third set of bus bars.

9. The device of claim 5 further comprising: a) each of the plurality of charge controllers and inverters being connected to each other in parallel; b) each of the plurality of charge controllers and inverters being electrically connected to one of a plurality of charge controller and inverter breaker panels, wherein there is one charge controller and inverter breaker panel for each charge controller and inverter; and c) each charge controller and inverter breaker panel being electrically connected to the third set of bus bars.

10. The device of claim 6 further comprising: a) each of the plurality of charge controllers and inverters being connected to each other in parallel; b) each of the plurality of charge controllers and inverters being electrically connected to one of a plurality of charge controller and inverter breaker panels, wherein there is one charge controller and inverter breaker panel for each charge controller and inverter; and c) each charge controller and inverter breaker panel being electrically connected to the third set of bus bars.

11. The device of claim 3 further comprising: a) each of the plurality of charge controllers and inverters being connected to each other in parallel; b) each of the plurality of charge controllers and inverters being electrically connected to one of a plurality of charge controller and inverter sub-panels contained in a main breaker box, wherein there is one charge controller and inverter breaker sub-panel for each charge controller and inverter; and c) each charge controller and inverter breaker sub-panel being electrically connected to the third set of bus bars.

12. The device of claim 4 further comprising: a) each of the plurality of charge controllers and inverters being connected to each other in parallel; b) each of the plurality of charge controllers and inverters being electrically connected to one of a plurality of charge controller and inverter sub-panels contained in a main breaker box, wherein there is one charge controller and inverter breaker

sub-panel for each charge controller and inverter; and c) each charger controller and inverter breaker sub-panel being electrically connected to the third set of bus bars.

13. The device of claim 5 further comprising: a) each of the plurality of charge controllers and inverters being connected to each other in parallel; b) each of the plurality of charge controllers and inverters being electrically connected to one of a plurality of charge controller and inverter sub-panels contained in a main breaker box, wherein there is one charge controller and inverter breaker sub-panel for each charge controller and inverter; and c) each charger controller and inverter breaker sub-panel being electrically connected to the third set of bus bars.

14. The device of claim 6 further comprising: a) each of the plurality of charge controllers and inverters being connected to each other in parallel; b) each of the plurality of charge controllers and inverters being electrically connected to one of a plurality of charge controller and inverter sub-panels contained in a main breaker box, wherein there is one charge controller and inverter breaker sub-panel for each charge controller and inverter; and c) each charger controller and inverter breaker sub-panel being electrically connected to the third set of bus bars.

15. The device of claim 11 further comprising: a) the first transfer switch being configured to transfer power to an electrical power grid when switched.

16. The device of claim 11 further comprising: a) the third transfer switch being configured to transfer power to an electrical power grid when switched.

17. The device of claim 3 further comprising: a) the third transfer switch being configured to transfer power to an electrical power grid when switched.

18. The device of claim 3 further comprising: a) the first transfer switch being configured to transfer power to an electrical power grid when switched.

19. The device of claim 7 further comprising: a) the third transfer switch being configured to transfer power to an electrical power grid when switched.

20. The device of claim 7 further comprising: a) the first transfer switch being configured to transfer power to an electrical power grid when switched.
